

Republic of the Philippines

Laguna State Polytechnic University

Province of Laguna

ITEP 414 - System Administration and Maintenance

BIOS vs UEFI Comparison and Modern Boot
Firmware Analysis

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Portfolio Project: Junior System Administrator Deployment Documentation

BIOS vs UEFI Comparison

Area	BIOS	UEFI
Definition	Legacy firmware interface used to initialize hardware and start the boot process.	Modern firmware interface that initializes hardware and launches an EFI boot application from firmware-managed boot entries.
Boot Process	Traditionally searches boot devices and transfers control to boot code, commonly associated with an MBR-based boot path.	Firmware can read the EFI System Partition and launch an EFI program such as a boot loader.
Maximum Disk Support	Legacy MBR booting is commonly limited to about 2 TiB on 512-byte-sector disks.	Normally paired with GPT, which supports far larger disks than MBR.
Partition Style	Commonly associated with MBR.	Normally uses GPT for modern installations.
Security Features	No standardized Secure Boot mechanism; security features depend heavily on platform implementation.	Supports Secure Boot and modern firmware security mechanisms when enabled and correctly configured.
Speed	Older initialization model; boot speed varies by hardware.	Can initialize hardware and boot entries more efficiently; actual boot speed still depends on hardware and configuration.
Advantages	Simple, widely understood on legacy systems, useful for older hardware and operating systems.	Modern disk support, flexible boot entries, Secure Boot support, graphical/extended firmware features, and better fit for current hardware.
Disadvantages	Legacy limitations, MBR constraints, fewer modern security and boot-management features.	More complex firmware settings; misconfigured Secure Boot or EFI entries can complicate troubleshooting.
Modern Usage	Mainly legacy compatibility and older systems.	Standard firmware approach on most modern PCs and servers.

Why UEFI Has Largely Replaced BIOS

UEFI has largely replaced traditional BIOS because modern computers need a firmware environment that can work with current storage sizes, flexible boot methods, stronger security options, and newer hardware designs. BIOS was created for an earlier generation of personal computers and was effective for many years, but its traditional boot model is closely associated with MBR limitations and legacy firmware conventions.

One major improvement is the way UEFI handles booting. Instead of relying only on a small boot record at the beginning of a disk, UEFI can use firmware boot entries and load an EFI executable from the EFI System Partition. This gives the firmware a clearer and more flexible method for selecting boot loaders. On a Linux server, an EFI-aware GRUB boot loader can then load the Linux kernel, after which `init/systemd` starts user-space services and eventually presents a login prompt.

UEFI is also commonly paired with the GUID Partition Table (GPT). GPT supports more partitions and much larger disks than the classic MBR scheme used with legacy BIOS booting. This is important in enterprise environments because servers may use large virtual or physical disks and may need storage layouts that are difficult to manage with older MBR limits.

Security is another reason for the transition. UEFI includes support for Secure Boot, which can help ensure that trusted boot components are used during startup. Secure Boot is not a complete security solution by itself, but it adds a firmware-level control that traditional BIOS did not standardize. This fits modern enterprise security practices, where administrators want protection to begin before the operating system fully starts.

UEFI also provides a more extensible firmware architecture. Vendors can implement richer configuration tools, network capabilities, firmware applications, and standardized interfaces that are difficult to provide through the traditional BIOS model. These features make UEFI better suited to current desktops, laptops, servers, and virtual machines.

BIOS knowledge is still valuable because older systems and compatibility modes remain in use, and many troubleshooting concepts are easier to understand when administrators know both approaches. However, for new

deployments, UEFI is generally preferred because it works better with modern storage, modern boot management, and modern security requirements. This is why understanding BIOS and UEFI is important for system administrators: firmware choices affect partitioning, boot-loader behavior, security controls, and the sequence that leads from power-on to a usable operating system.

References

- Laguna State Polytechnic University, ITEP 414 Week 3 Portfolio Project.
- UEFI Forum specifications and documentation.
- Official operating-system documentation for UEFI, GPT, Secure Boot, and boot-loader behavior.